SRI CHANDRASEKHARENDRA SARASWATHI VISWA MAHAVIDYALAYA

(UNIVERSITY ESTABLISHED UNDER SECTION 3 OF UGC ACT 1956)
ENATHUR, KANCHIPURAM – 631 561

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Class : II B.E. (CSE)

Course Code: BCSF244AI2

Course Name:Data Science and Toolkit Lab

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BONAFIDE CERTIFICATE

This is to certify that this is the bonafide record of the **Data Science Toolkit Lab** with the subject code **BCSF244AI2** was done by

Mr/Ms. Pradhamesh M ,

with **Reg. No 11249A296** of **II-B.E.(CSE)** during the academic year 2025 – 2026.

Station:Enathur

Date:

Staff-in-charge

Head of the Department

Submitted for the Practical examination held on_____.

Examiner-1

Examiner-2

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Exp 1: Introduction to python libraries

Date : 29-01-2026

1.1 NumPy

Aim: To implement array creation and operations using NumPy.

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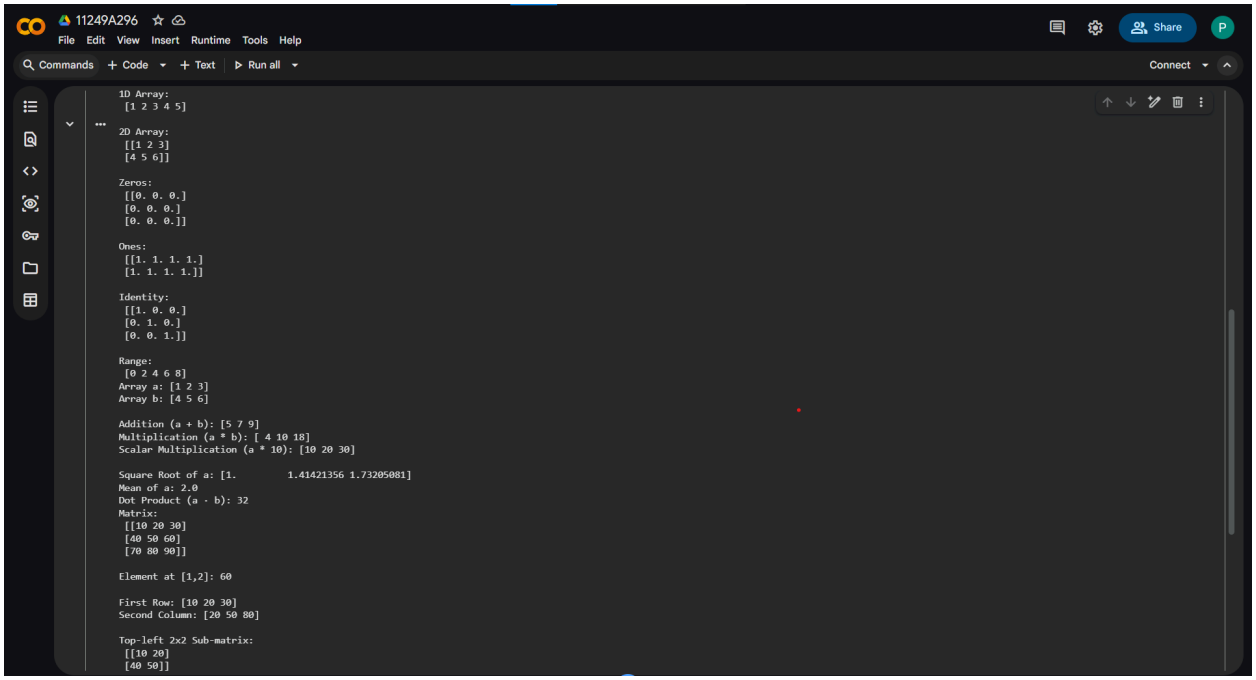
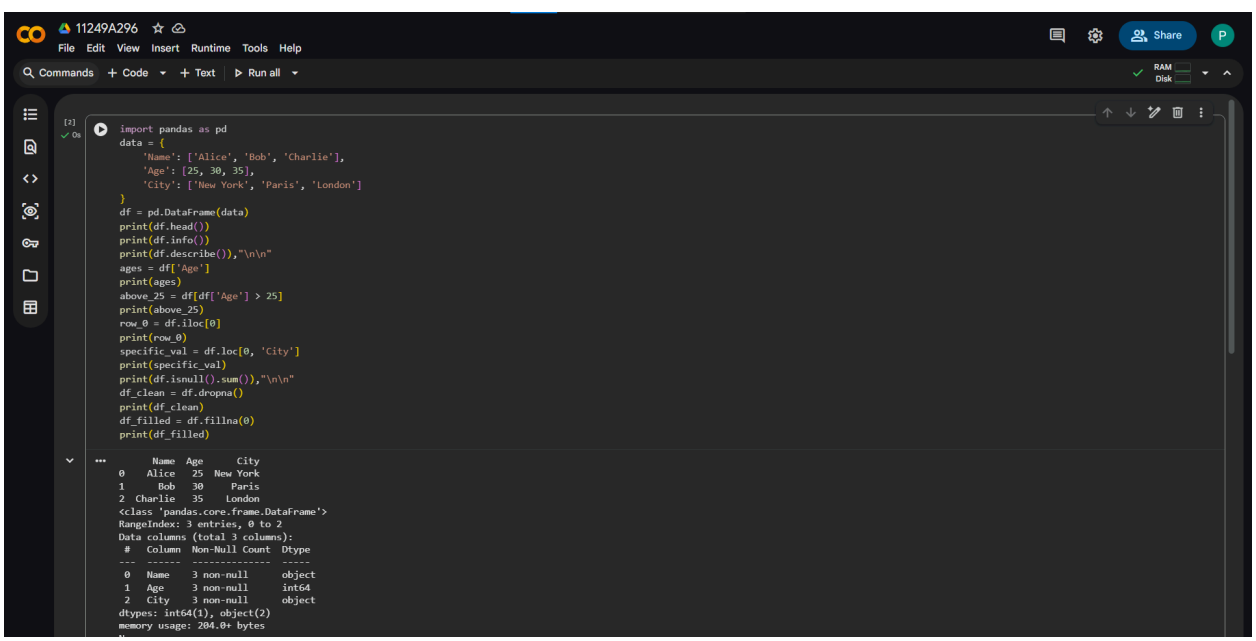
```python
import numpy as np
print(

"1D Array:\n", np.array([1,2,3,4,5]), "\n\n"
"2D Array:\n", np.array([[1,2,3],[4,5,6]]), "\n\n"
"Zeros:\n", np.zeros((3,3)), "\n\n"
"Ones:\n", np.ones((2,4)), "\n\n"
"Identity:\n", np.eye(3), "\n\n"
"Range:\n", np.arange(0,10,2)
)

import numpy as np
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print(
"Array a:", a, "\n"
"Array b:", b, "\n\n"
"Addition (a + b):", a + b, "\n"
"Multiplication (a * b):", a * b, "\n"
"Scalar Multiplication (a * 10):", a * 10, "\n\n"
"Square Root of a:", np.sqrt(a), "\n"
"Mean of a:", np.mean(a), "\n"
"Dot Product (a · b):", np.dot(a, b)
)

import numpy as np
matrix = np.array([[10, 20, 30],
[40, 50, 60],
[70, 80, 90]])
print(
"Matrix:\n", matrix, "\n\n"
"Element at [1,2]:", matrix[1, 2], "\n\n"
"First Row:", matrix[0, :], "\n"
"Second Column:", matrix[:, 1], "\n\n"
"Top-left 2x2 Sub-matrix:\n", matrix[0:2, 0:2]
)
```



```
1D Array:
 [1 2 3 4 5]

2D Array:
 [[1 2 3]
 [4 5 6]]

Zeros:
 [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

Ones:
 [[1. 1. 1. 1.]
 [1. 1. 1. 1.]]

Identity:
 [[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]

Range:
 [0 2 4 6 8]
Array a: [1 2 3]
Array b: [4 5 6]

Addition (a + b): [5 7 9]
Multiplication (a * b): [ 4 10 18]
Scalar Multiplication (a * 10): [10 20 30]

Square Root of a: [1.         1.41421356 1.73205081]
Mean of a: 2.0
Dot Product (a · b): 32
Matrix:
 [[10 20 30]
 [40 50 60]
 [70 80 90]]

Element at [1,2]: 60

First Row: [10 20 30]
Second Column: [20 50 80]

Top-left 2x2 Sub-matrix:
 [[10 20]
 [40 50]]
```

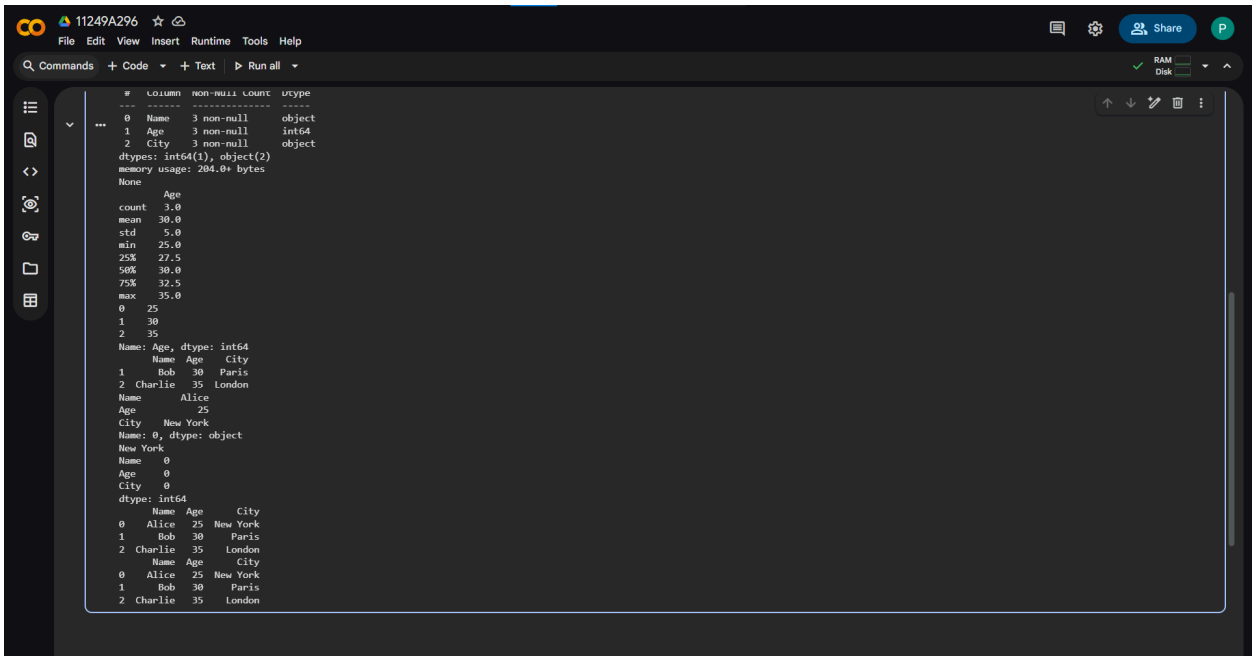
1.2 Pandas

Aim: To implement data creations, filtering, and cleaning using Pandas



```
import pandas as pd
data = {
    'Name': ['Alice', 'Bob', 'Charlie'],
    'Age': [25, 30, 35],
    'City': ['New York', 'Paris', 'London']
}
df = pd.DataFrame(data)
print(df.head())
print(df.info())
print(df.describe()),"\n\n"
ages = df['Age']
print(ages)
above_25 = df[df['Age'] > 25]
print(above_25)
row_0 = df.iloc[0]
print(row_0)
specific_val = df.loc[0, 'City']
print(specific_val)
print(df.isnull().sum()),"\n\n"
df_clean = df.dropna()
print(df_clean)
df_filled = df.fillna(0)
print(df_filled)
```

```
      Name  Age      City
0    Alice   25  New York
1      Bob   30     Paris
2  Charlie   35    London
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3 entries, 0 to 2
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Name    3 non-null      object
 1   Age     3 non-null      int64
 2   City    3 non-null      object
dtypes: int64(1), object(2)
memory usage: 204.0+ bytes
```



1.3 Matplot lib

Aim: To implement plotting operations using Matplolib.

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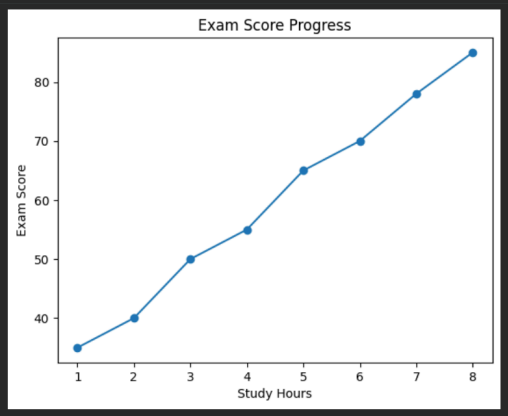
```
plt.plot (df['Study Hours'],df['Exam Score'],marker='o')
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.title('Exam Score Progress')
plt.show()
```



Variables Terminal 5:31 PM Python 3

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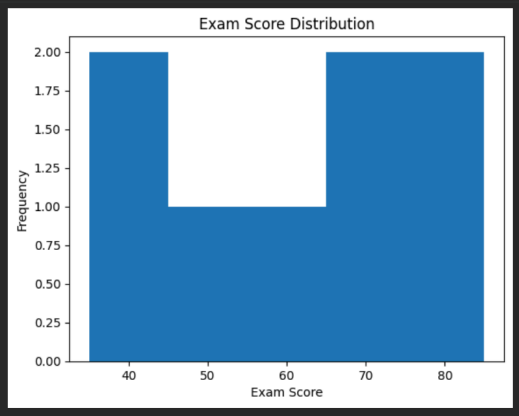
Commands + Code + Text Run all

```
plt.hist(df['Exam Score'],bins=5)
plt.xlabel('Exam Score')
plt.ylabel('Frequency')
plt.title('Exam Score Distribution')
plt.show()
```



Variables Terminal 5:31 PM Python 3

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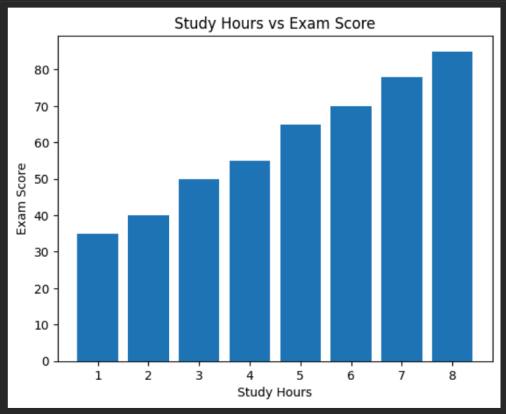
Commands + Code + Text Run all

```
plt.bar (df['Study Hours'],df['Exam Score'])
plt.xlabel('Study Hours')
plt.ylabel('Exam Score')
plt.title('Study Hours vs Exam Score')
plt.show()
```



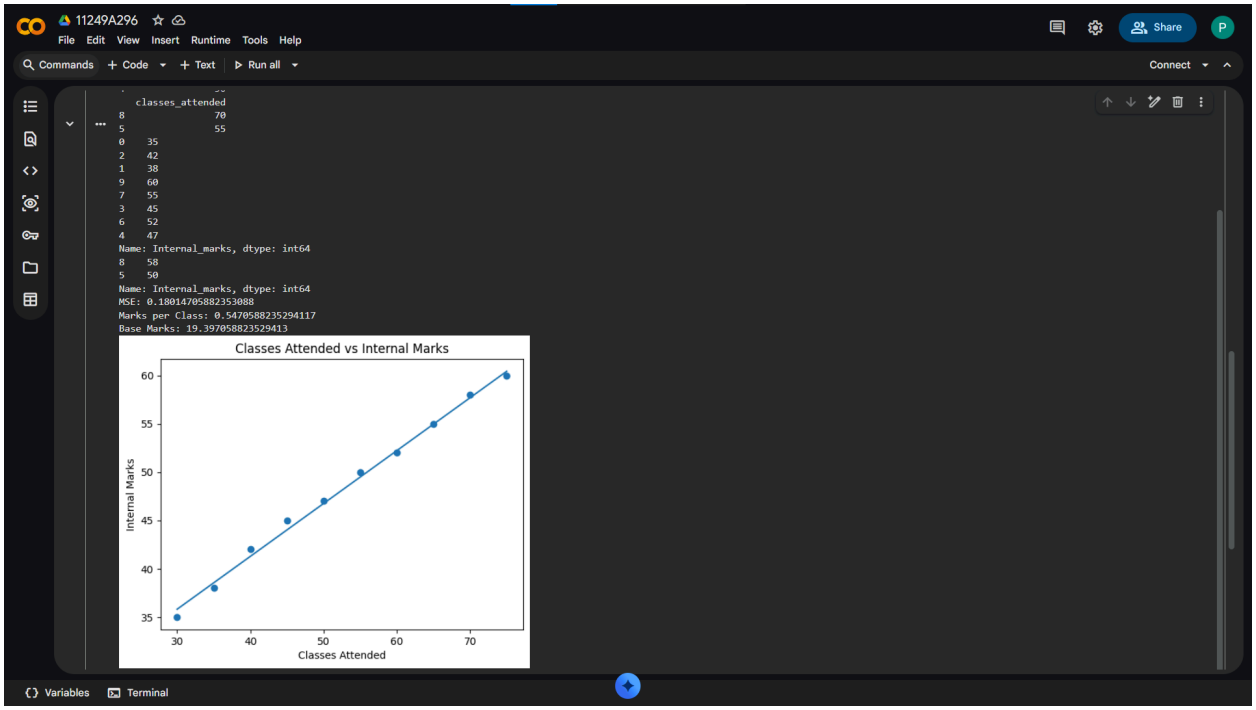
Variables Terminal 5:31 PM Python 3

1.4 Scikit-Learn

Aim: To implement machine learning algorithms using Scikit-learn.

```python
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error
data = {
    'Student' : ['S1','S2','S3','S4','S5','S6','S7','S8','S9','S10'],
    'classes_attended' : [30,35,40,45,50,55,60,65,70,75],
    'Internal_marks' : [35,38,42,45,47,50,52,55,58,60],
}
df = pd.Dataframe(data)
print(df)

X=df[['classes_attended']]
print(X)
y=df['Internal_marks']
print(y)

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=7)
print(X_train)
print(X_test)
print(y_train)
print(y_test)

model = LinearRegression()
model.fit(X_train,y_train)

predictions = model.predict(X_test)

mse = mean_squared_error(y_test, predictions)
print("MSE:", mse)
print("Marks per Class:", model.coef_[0])
print("Base Marks:", model.intercept_)

plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Classes Attended")
plt.ylabel("Internal Marks")
plt.title("Classes Attended vs Internal Marks")
plt.show()
```

```
  Student  classes_attended  Internal_marks
0      S1                30              35
1      S2                35              38
2      S3                40              42
3      S4                45              45
4      S5                50              47
5      S6                55              50
6      S7                60              52
7      S8                65              55
8      S9                70              58
9     S10                75              60
   classes_attended
0                30
1                35
2                40
3                45
4                50
5                55
6                60
7                65
8                70
9                75
0    35
1    38
2    42
3    45
4    47
5    50
6    52
7    55
8    58
9    60
Name: Internal_marks, dtype: int64
   classes_attended
0                30
2                40
1                35
9                75
7                65
3                45
6                60
4                50
   classes_attended
```



Result: Machine learning models were implemented successfully.

Exp 2 : Data Exploration and preprocessing in python
Date: 13-02-2026

Aim: To implement data exploration and preprocessing.

```python
import pandas as pd
import numpy as np
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import train_test_split

data = {
'Country': ['India', 'USA', 'India', 'USA', 'UK', 'India'],
'Age': [22, 25, np.nan, 30, 28, 35],
'Salary': [40000, np.nan, 50000, np.nan, 72000, 58000],
'Purchased': ['No', 'Yes', 'Yes', 'No', 'Yes', 'Yes']
}
df = pd.DataFrame(data)
print("--- ORIGINAL RAW DATA ---")
print(df)
print("\n")

print("--- MISSING VALUES COUNT ---")
print(df.isnull().sum())
print("\n")
df['Age'] = df['Age'].fillna(df['Age'].mode())
df['Salary'] = df['Salary'].fillna(df['Salary'].mode())

print("--- DATA AFTER CLEANING ---")
print(df)
print("\n")
df_encoded = pd.get_dummies(df, columns=['Country'])
df_encoded['Purchased'] = df_encoded['Purchased'].map({'Yes': 1, 'No': 0})

print("--- DATA AFTER ENCODING (All Numbers Now) ---")
print(df_encoded)
print("\n")

X = df_encoded.drop('Purchased', axis=1)
y = df_encoded['Purchased']

scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print("--- FINAL PROCESSED DATA (Ready for AI) ---")
print(pd.DataFrame(X_scaled, columns=X.columns).head())
```

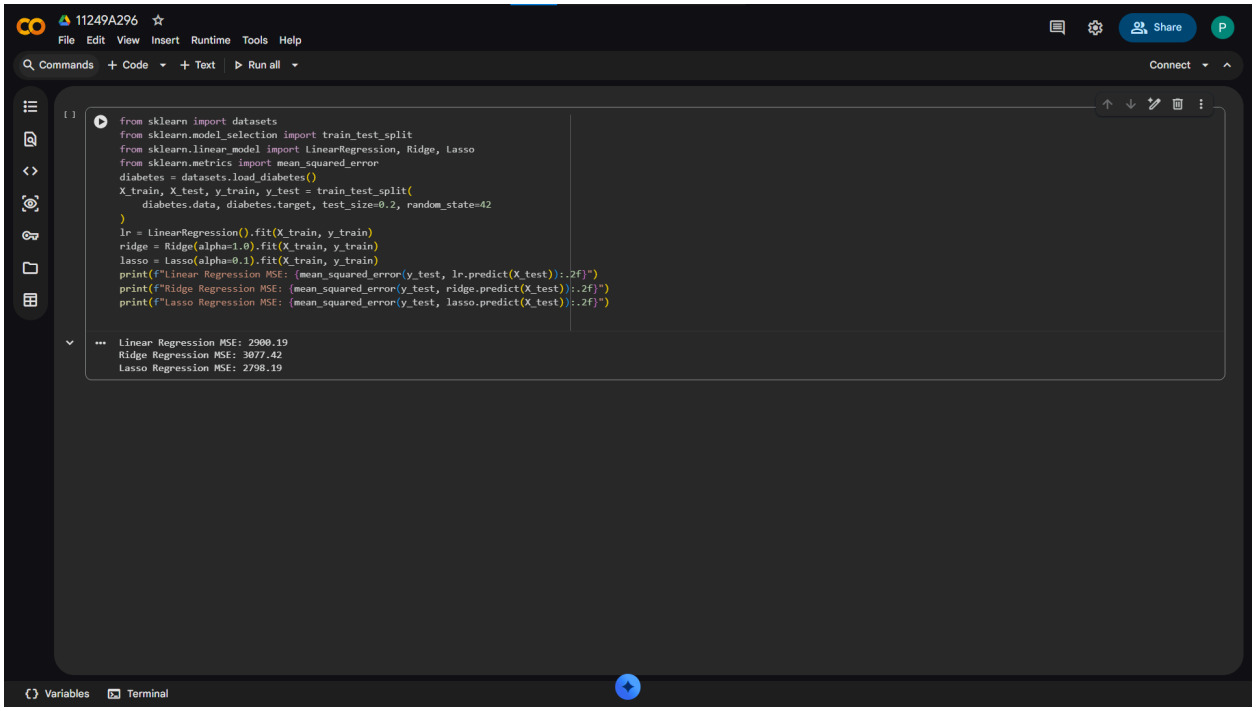
```
--- ORIGINAL RAW DATA ---
  Country   Age   Salary Purchased
0   India  22.0  40000.0        No
1     USA  25.0      NaN       Yes
2   India   NaN  50000.0       Yes
3     USA  30.0      NaN        No
4      UK  28.0  72000.0       Yes
5   India  35.0  58000.0       Yes


--- MISSING VALUES COUNT ---
Country      0
Age          1
Salary       2
Purchased    0
dtype: int64


--- DATA AFTER CLEANING ---
  Country   Age   Salary Purchased
0   India  22.0  40000.0        No
1     USA  25.0  50000.0       Yes
2   India  28.0  50000.0       Yes
3     USA  30.0  72000.0        No
4      UK  28.0  72000.0       Yes
5   India  35.0  58000.0       Yes


--- DATA AFTER ENCODING (All Numbers Now) ---
   Age   Salary  Purchased  Country_India  Country_UK  Country_USA
0  22.0  40000.0          0           True       False        False
1  25.0  50000.0          1          False       False         True
2  28.0  50000.0          1           True       False        False
3  30.0  72000.0          0          False       False         True
4  28.0  72000.0          1          False        True        False
5  35.0  58000.0          1           True       False        False


--- FINAL PROCESSED DATA (Ready for AI) ---
        Age    Salary  Country_India  Country_UK  Country_USA
0 -1.484615 -1.438476            1.0   -0.447214    -0.707107
1 -0.742307 -0.592314           -1.0   -0.447214     1.414214
2  0.000000 -0.592314            1.0   -0.447214    -0.707107
3  0.494872  1.269243           -1.0   -0.447214     1.414214
4  0.000000  1.269243           -1.0    2.236068    -0.707107
```

Result : Data preprocessing was performed successfully.

Exp 3 : Implement Regularized Linear Regression

Date : 19-02-2026

Aim: To implement regularized linear regression.

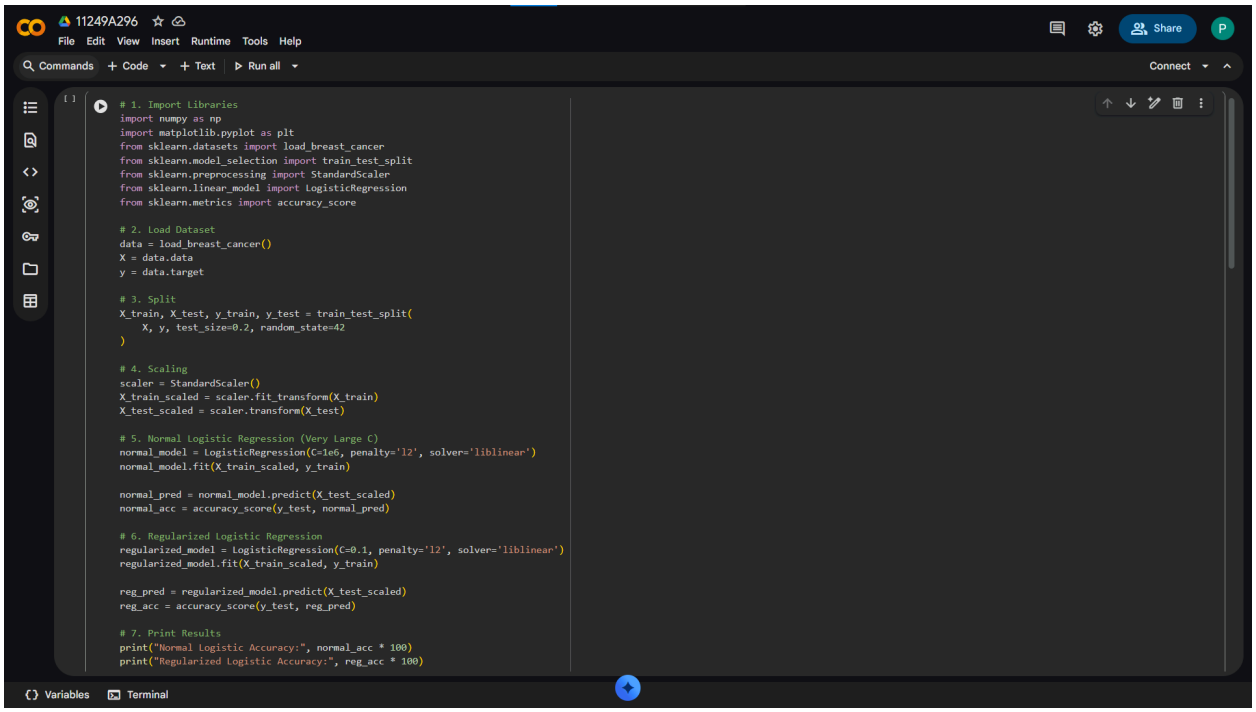


Result : The regression model was implemented successfully

Exp 4 : Implement Naive Bayes Classifier

Date : 05-03-2026

Aim: To implement Navie Bayes Classification.



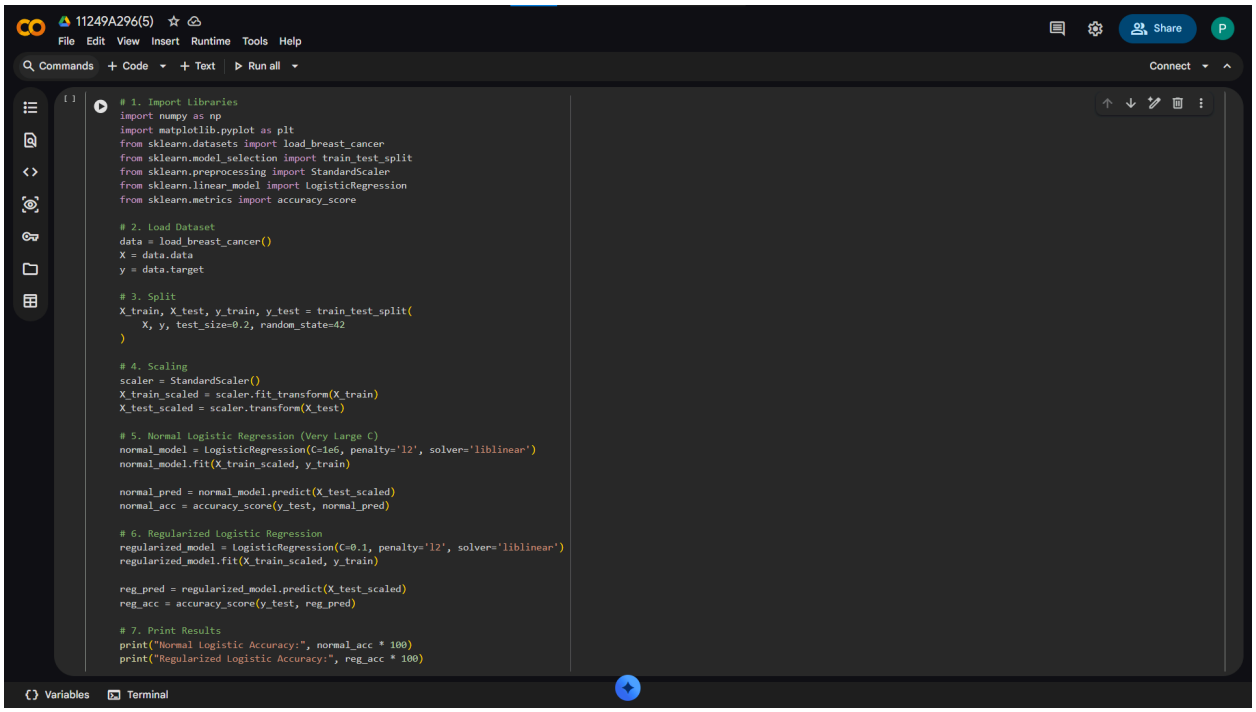
```python
# 1. Import Libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# 2. Load Dataset
data = load_breast_cancer()
X = data.data
y = data.target

# 3. Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 4. Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 5. Normal Logistic Regression (Very Large C)
normal_model = LogisticRegression(C=1e6, penalty='l2', solver='liblinear')
normal_model.fit(X_train_scaled, y_train)

normal_pred = normal_model.predict(X_test_scaled)
normal_acc = accuracy_score(y_test, normal_pred)

# 6. Regularized Logistic Regression
regularized_model = LogisticRegression(C=0.1, penalty='l2', solver='liblinear')
regularized_model.fit(X_train_scaled, y_train)

reg_pred = regularized_model.predict(X_test_scaled)
reg_acc = accuracy_score(y_test, reg_pred)

# 7. Print Results
print("Normal Logistic Accuracy:", normal_acc * 100)
print("Regularized Logistic Accuracy:", reg_acc * 100)
```

Result: The Classifier was implemented successfully.

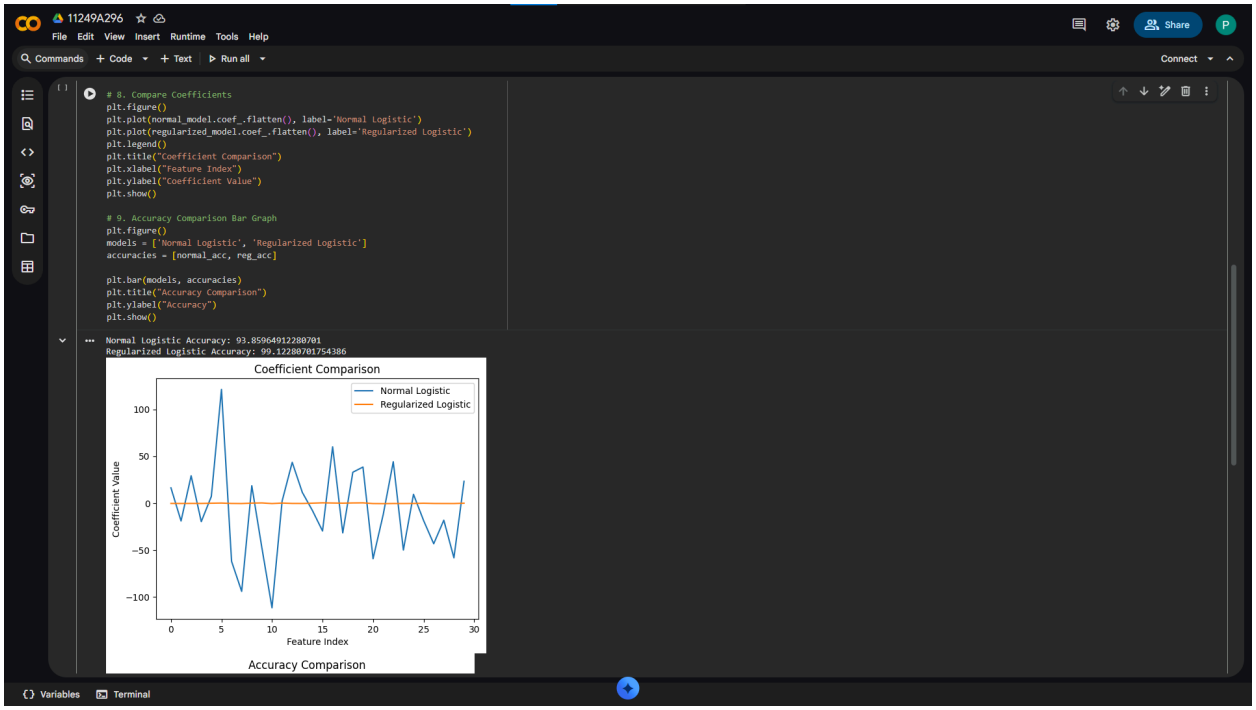
Exp 5:Implement Regularised logistic regression

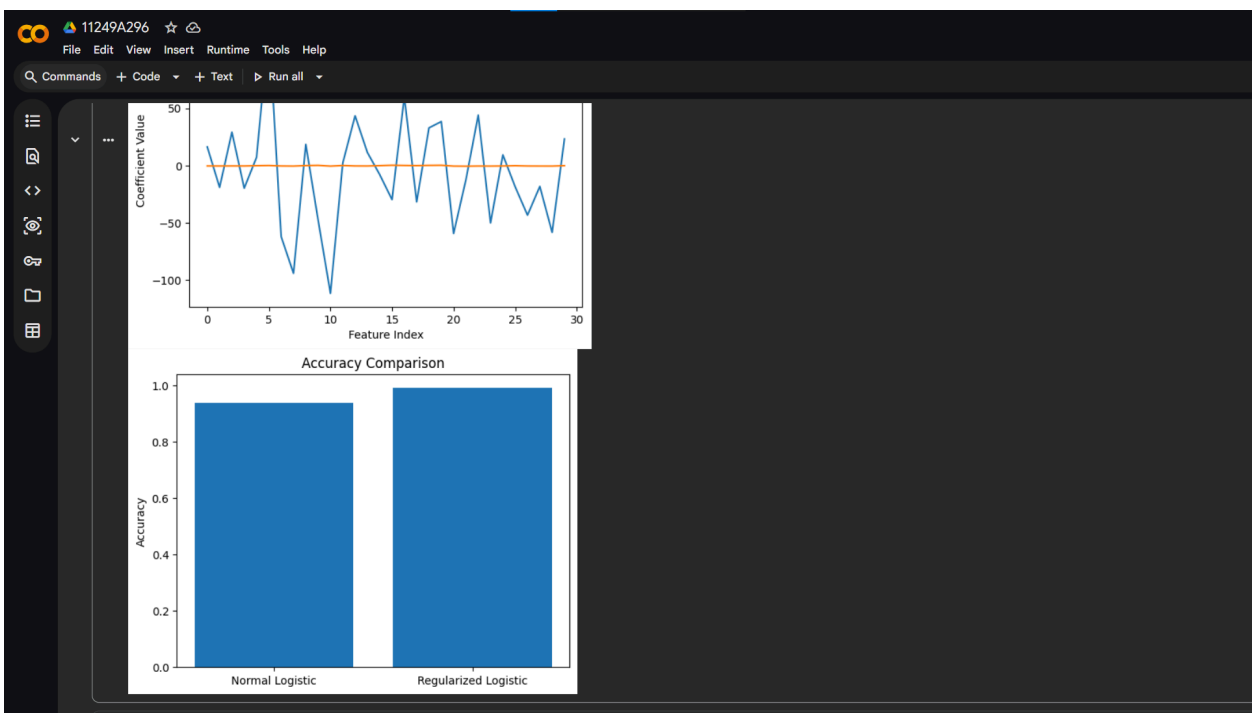
Date : 05-03-2026

Aim: To implement regularized logistic regression.



```python
# 1. Import Libraries
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

# 2. Load Dataset
data = load_breast_cancer()
X = data.data
y = data.target

# 3. Split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# 4. Scaling
scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

# 5. Normal Logistic Regression (Very Large C)
normal_model = LogisticRegression(C=1e6, penalty='l2', solver='liblinear')
normal_model.fit(X_train_scaled, y_train)

normal_pred = normal_model.predict(X_test_scaled)
normal_acc = accuracy_score(y_test, normal_pred)

# 6. Regularized Logistic Regression
regularized_model = LogisticRegression(C=0.1, penalty='l2', solver='liblinear')
regularized_model.fit(X_train_scaled, y_train)

reg_pred = regularized_model.predict(X_test_scaled)
reg_acc = accuracy_score(y_test, reg_pred)

# 7. Print Results
print("Normal Logistic Accuracy:", normal_acc * 100)
print("Regularized Logistic Accuracy:", reg_acc * 100)
```



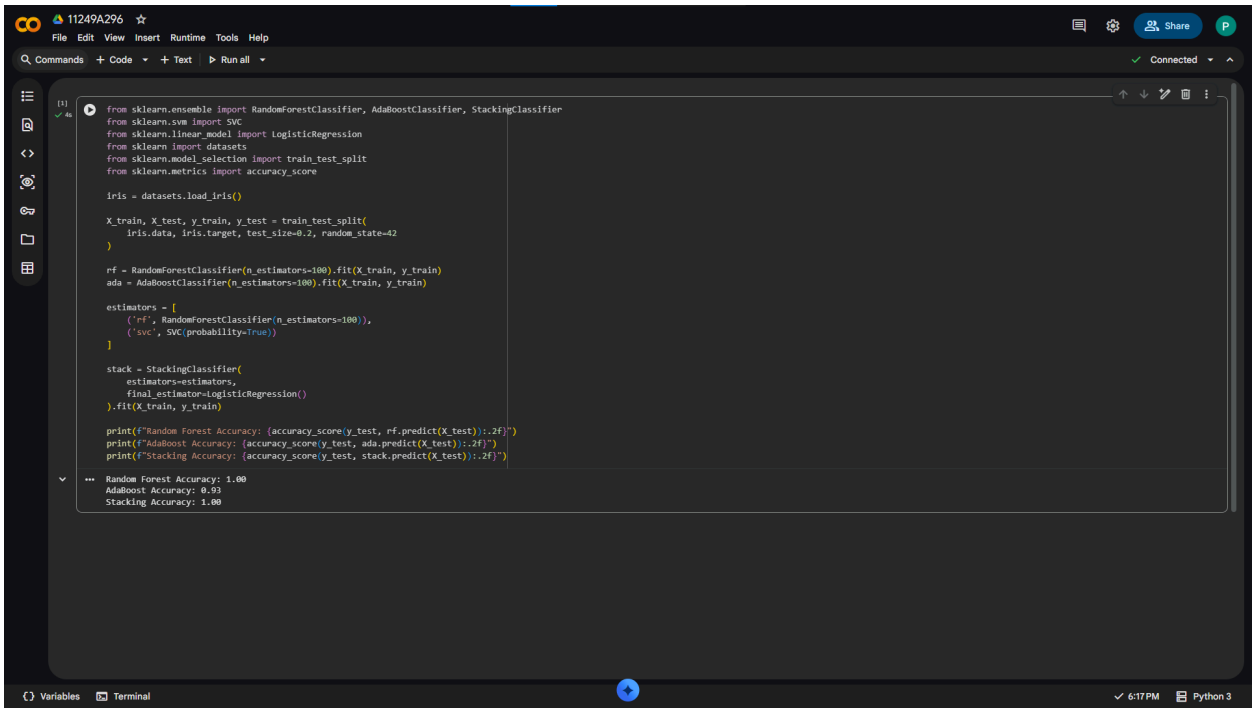


Result: The logistic regression model was implemented successfully.

Exp 6 : Build Models using different Ensembling Techniques

Date : 12-03-2026

Aim: To implement ensembling techniques.



```python
from sklearn.ensemble import RandomForestClassifier, AdaBoostClassifier, StackingClassifier
from sklearn.svm import SVC
from sklearn.linear_model import LogisticRegression
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

iris = datasets.load_iris()

X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42
)

rf = RandomForestClassifier(n_estimators=100).fit(X_train, y_train)
ada = AdaBoostClassifier(n_estimators=100).fit(X_train, y_train)

estimators = [
    ('rf', RandomForestClassifier(n_estimators=100)),
    ('svc', SVC(probability=True))
]

stack = StackingClassifier(
    estimators=estimators,
    final_estimator=LogisticRegression()
).fit(X_train, y_train)

print(f"Random Forest Accuracy: {accuracy_score(y_test, rf.predict(X_test)):.2f}")
print(f"AdaBoost Accuracy: {accuracy_score(y_test, ada.predict(X_test)):.2f}")
print(f"Stacking Accuracy: {accuracy_score(y_test, stack.predict(X_test)):.2f}")
```

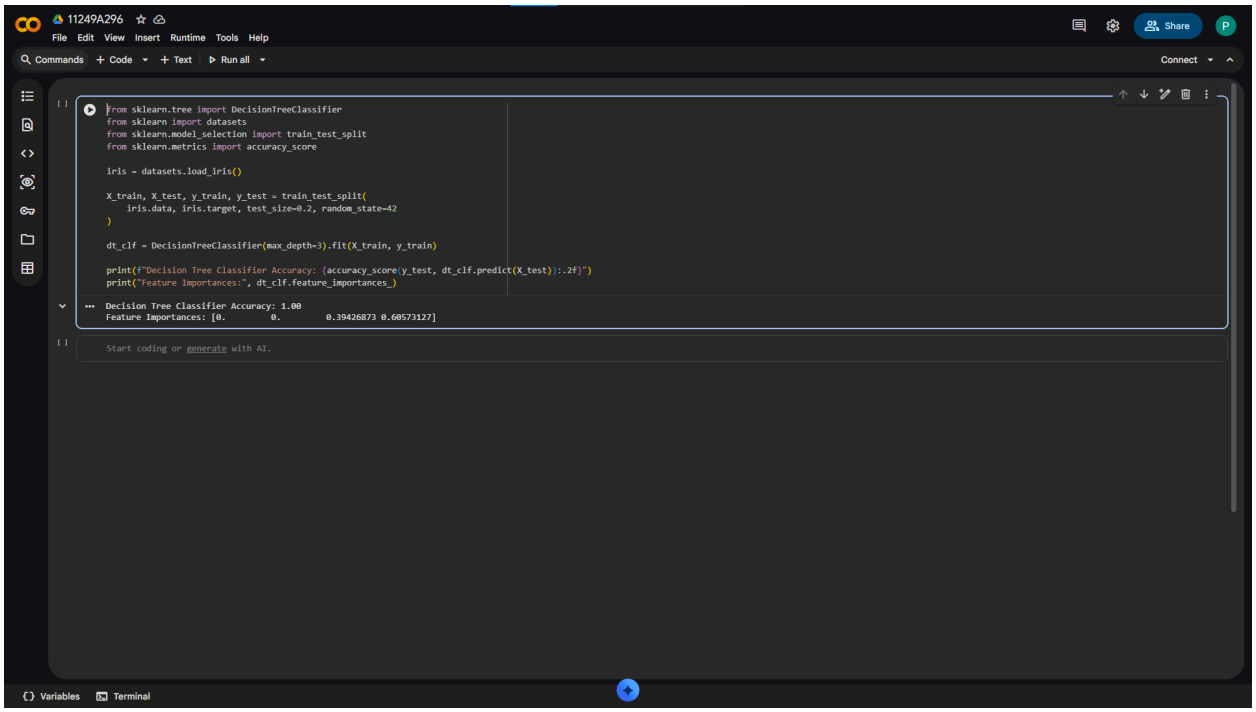
```
Random Forest Accuracy: 1.00
AdaBoost Accuracy: 0.93
Stacking Accuracy: 1.00
```

Result: The ensembling techniques were implemented successfully.

Exp 7 : Build Models using Decision Trees

Date : 12-03-2026

Aim: To implement decision tree models.



```
from sklearn.tree import DecisionTreeClassifier
from sklearn import datasets
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

iris = datasets.load_iris()

X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42
)

dt_clf = DecisionTreeClassifier(max_depth=3).fit(X_train, y_train)

print(f"Decision Tree Classifier Accuracy: {accuracy_score(y_test, dt_clf.predict(X_test)):.2f}")
print("Feature Importances:", dt_clf.feature_importances_)
```

```
Decision Tree Classifier Accuracy: 1.00
Feature Importances: [0.         0.         0.39426873 0.60573127]
```

Result: The decision tree models were implemented successfully.

Exp 8: Support Vector Machines (SVM) with various Kernels

Date : 02-04-2026

Aim: To implement Support Vector Machine.

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```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score
from sklearn.datasets import load_iris

data = load_iris()

X = data.data
y = data.target

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

model_linear = SVC(kernel='linear')
model_linear.fit(X_train, y_train)

pred_linear = model_linear.predict(X_test)

acc_linear = accuracy_score(y_test, pred_linear)

print("Linear Kernel Accuracy:", acc_linear)

model_poly = SVC(kernel='poly', degree=3)
model_poly.fit(X_train, y_train)

pred_poly = model_poly.predict(X_test)
acc_poly = accuracy_score(y_test, pred_poly)

print("Polynomial Kernel Accuracy:", acc_poly)

model_rbf = SVC(kernel='rbf')
model_rbf.fit(X_train, y_train)

pred_rbf = model_rbf.predict(X_test)
acc_rbf = accuracy_score(y_test, pred_rbf)

print("RBF Kernel Accuracy:", acc_rbf)
```

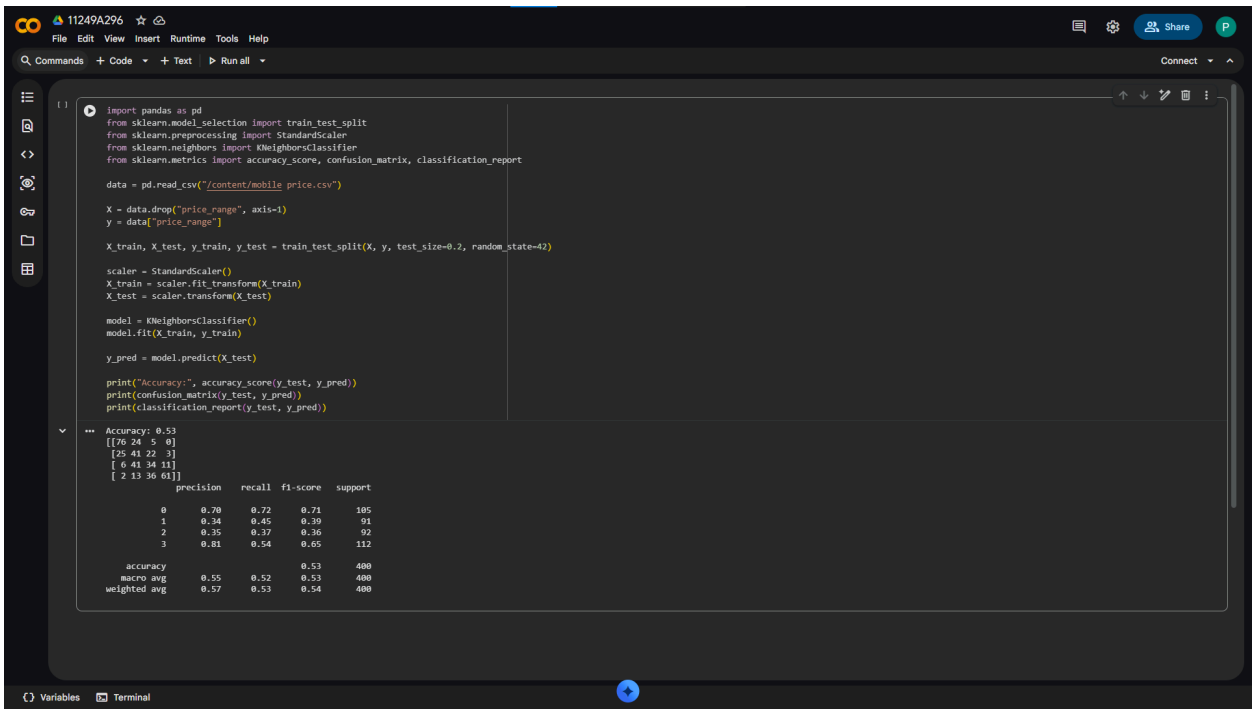
```
Linear Kernel Accuracy: 1.0
Polynomial Kernel Accuracy: 1.0
RBF Kernel Accuracy: 1.0
```

Result: The Support Vector Machines with Kernels was implemented successfully

Exp 9:-Implement KNN Classification

Date : 02-04-2026

Aim: To implement K-Nearest Neighbors algorithm



```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

data = pd.read_csv("/content/mobile price.csv")

X = data.drop("price_range", axis=1)
y = data["price_range"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

model = KNeighborsClassifier()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

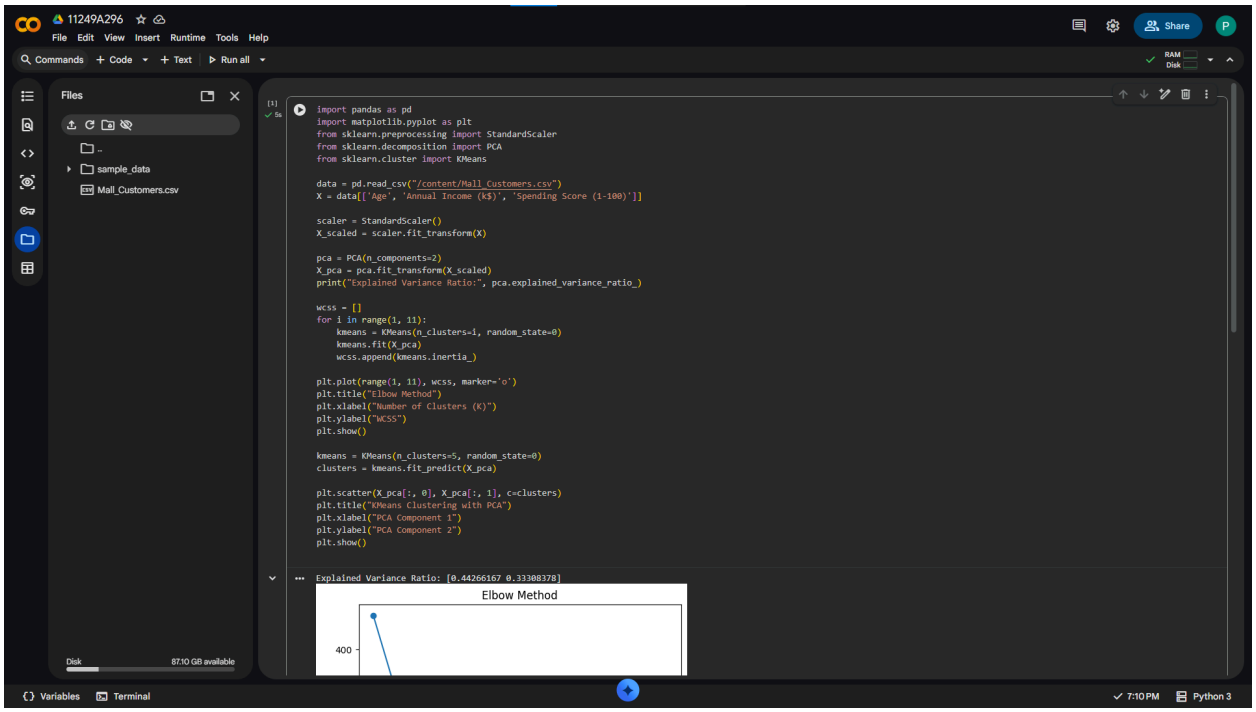
```
Accuracy: 0.53
[[76 24  5  0]
 [25 41 22  3]
 [ 6 41 34 11]
 [ 2 13 36 61]]
              precision    recall  f1-score   support

           0       0.70      0.72      0.71       105
           1       0.34      0.45      0.39        91
           2       0.35      0.37      0.36        92
           3       0.81      0.54      0.65       112

    accuracy                           0.53       400
   macro avg       0.55      0.52      0.53       400
weighted avg       0.57      0.53      0.54       400
```

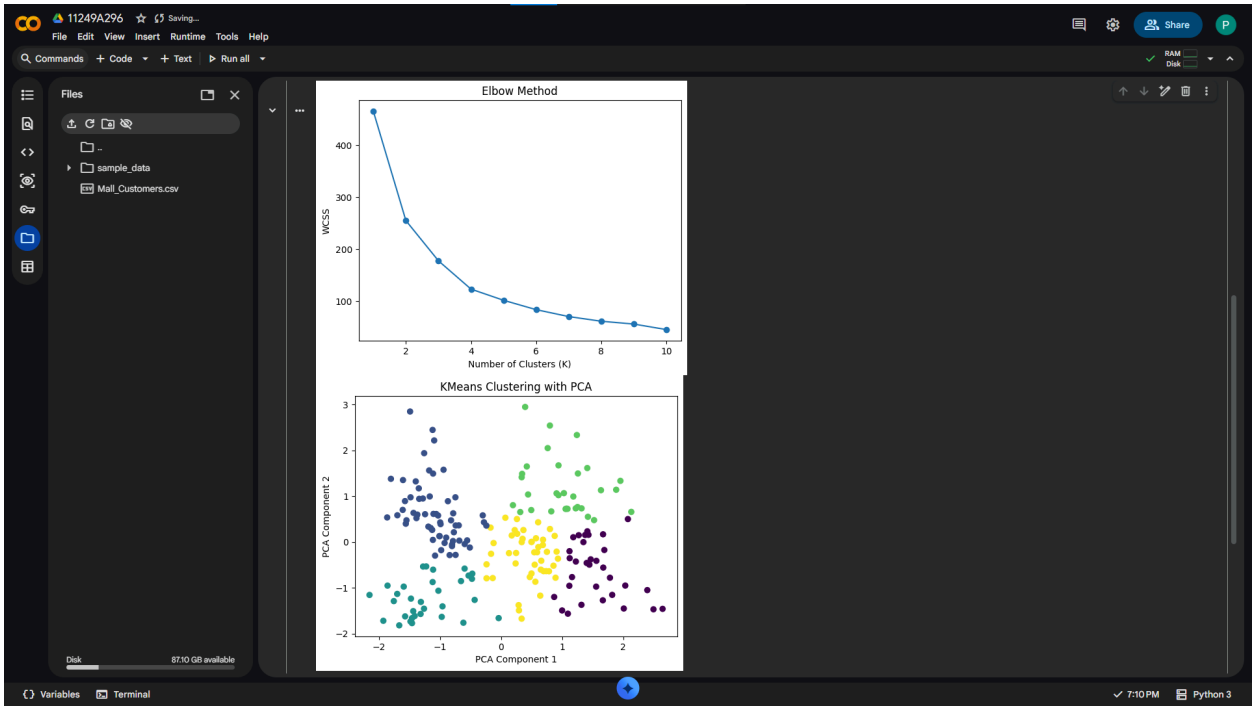
Result: The KNN Classification model was implemented successfully.

Exp 10 : K- Means Clustering with PCA and Elbow Method

Date : 02-04-2026

Aim: To implement k-means clustering.





Result: K- Means Clustering with PCA and Elbow Method was performed successfully.